

CDS503 Machine Learning

Assignment 2

Report

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Part 1: K-Means Clustering

- a) For k-means clustering, number of clusters of 2, 3, 4, 5, 6, 7, 8 and 9 are tried with the random state of 46 and the sum of squared distances to centroids are observed. Table 1 reports the sum of squared distances to centroids for every number of clusters rounded to the nearest integer.

Table 1 Sum of squared distances to centroids for different number of clusters for k-means clustering

Number of clusters	Sum of squared distances to centroids	Decrease in sum of squared distances to centroids
2	725407	-
3	356366	369041
4	288531	67835
5	242939	45592
6	214291	28648
7	131838	82453
8	113389	18449
9	108131	5258

As shown in Table 1, the sum of squared distances to centroids decreases as the number of clusters increases. This shows that the clustering is better as the number of clusters increases. Since the number of squared distances to centroids shows little decrease between number of clusters of 7 and 8, this indicates that 7 is a good number of clusters. So, the number of clusters that best clusters the countries into continents is 7.

- b) The parameter tested are random_state of 0, 23, 79, 46, 457, 326, 5984, 48934, 100, 32, 119, 6, 298, 5901 and 53, and algorithm of lloyd and elkan. Different random states are tested because k-means clustering is sensitive to initial seeds. The results show that random_state of 326 gives the lowest sum of squared distances to centroids of 131830. For algorithm of lloyd and elkan, the sum of squared distances to centroids are the same, so there is no way to differentiate which algorithm is better. Since neither of these two algorithms is better than another, lloyd is chosen for the reason that it is not worse than elkan. The following are the final parameters chosen to obtain my final clusters:
- n_clusters (k): 7
 - random_state: 326
 - algorithm: lloyd

The coordinates of the cluster centroids are listed below:

- Cluster 0: (latitude: -12.081357, longitude: 31.317325)
- Cluster 1: (latitude: 9.620036, longitude: -69.149071)
- Cluster 2: (latitude: 39.545329, longitude: 6.285904)
- Cluster 3: (latitude: -10.813535, longitude: 156.117259)
- Cluster 4: (latitude: -15.842108, longitude: -164.351001)
- Cluster 5: (latitude: 31.800491, longitude: 48.654594)
- Cluster 6: (latitude: 18.348466, longitude: 103.455094)

The number of iterations is 5. The sum of squared differences to centroid is 131830. I think this is the best clustering because the sum of squared distances to centroids is low.

c) Table 2 lists the centroid and continent label for each cluster.

Table 2 Continents for each cluster (k-means)

Cluster	Centroid	Continent
0	latitude: -12.081357, longitude: 31.317325	Africa
1	latitude: 9.620036, longitude: -69.149071	Americas (North America and South America)
2	latitude: 39.545329, longitude: 6.285904	Europe
3	latitude: -10.813535, longitude: 156.117259	Oceania (Australia and Pacific) and Antarctica
4	latitude: -15.842108, longitude: -164.351001	Oceania (remote Pacific Islands)
5	latitude: 31.800491, longitude: 48.654594	Asia (Middle East)
6	latitude: 18.348466, longitude: 103.455094	Asia (East Asia, South Asia and Southeast Asia)

Figure 1 is the scatter plot of the best k-means clustering.

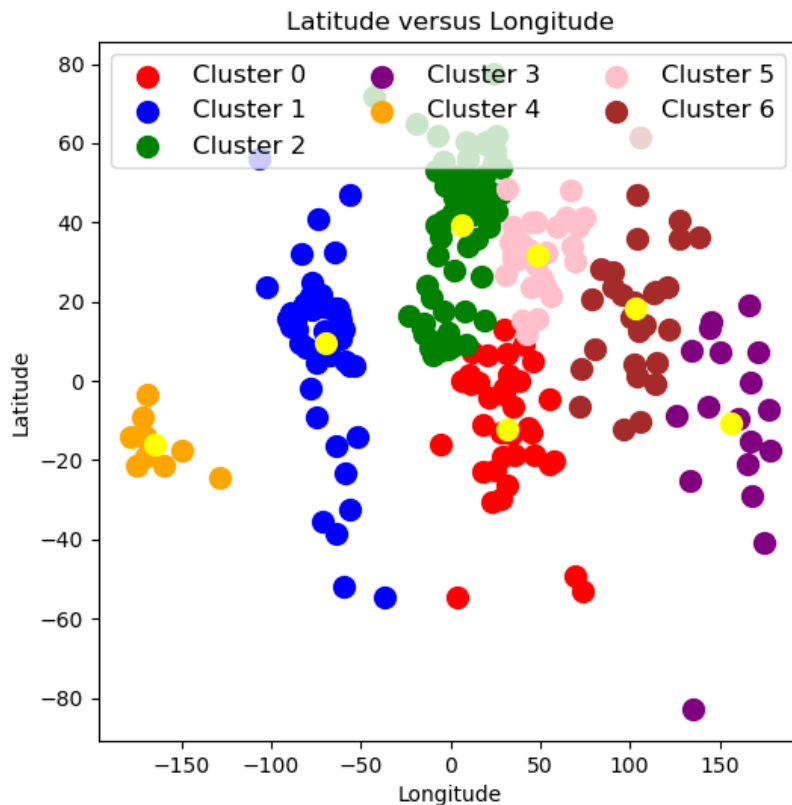


Figure 1 Scatter plot of k-means clustering

Part 2: Hierarchical Clustering

a) For the hierarchical clustering (agglomerative clustering), single linkage, complete linkage and average linkage are tried and the dendrogram for each linkage method is drawn. The best parameter and number of clusters that I have selected are:

- method / linkage: complete
- n_clusters: 7

Figure 2 is the dendrogram when the linkage is complete linkage, whereas Figure 3 is the scatter plot that shows the 7 clusters as a result of agglomerative clustering using complete linkage.

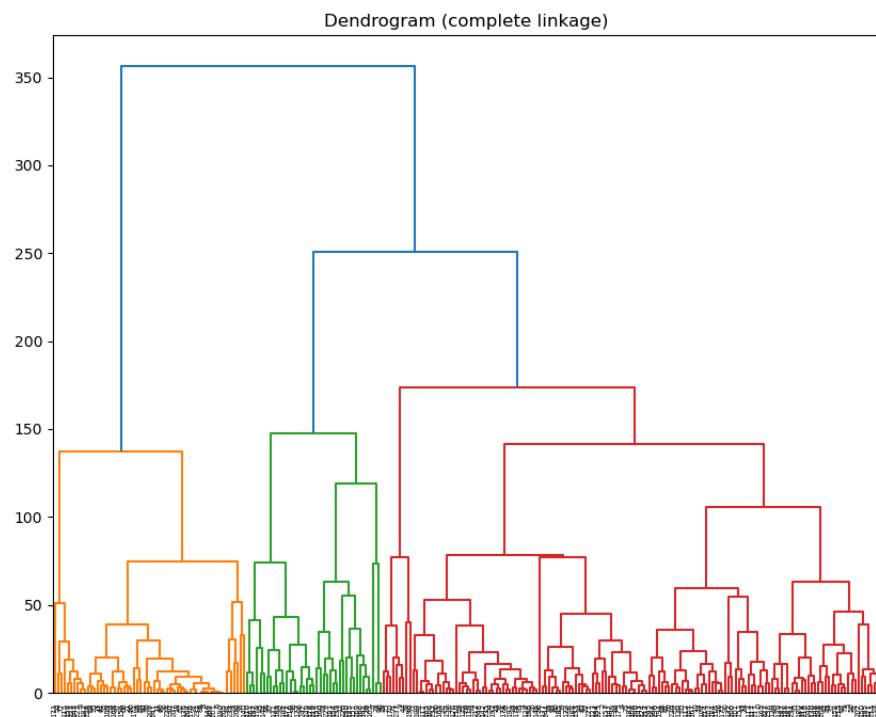


Figure 2 Dendrogram of complete linkage

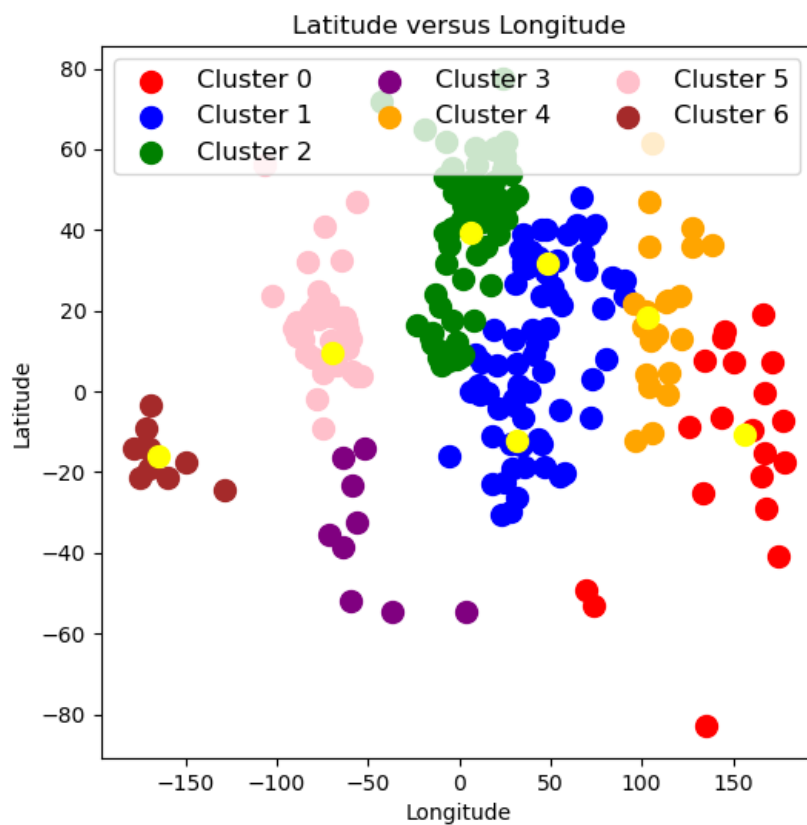


Figure 3 Scatter plot of hierarchical clustering (agglomerative clustering)

I think that the number of clusters and linkage method that I have selected results in the best clustering because the size of each cluster is more balanced compared to other parameters.

b) Table 3 lists the continent label for each cluster.

Table 3 Continents for each cluster (hierarchical clustering / agglomerative clustering)

Cluster	Continent
0	Oceania (Australia and pacific) and Antarctica
1	Africa and Asia (Middle East)
2	Europe
3	South America
4	Asia (East Asia, South Asia and Southeast Asia)
5	North America
6	Oceania (remote Pacific islands)

Part 3: Identify the best continent clustering

a) Among k-means clustering and hierarchical clustering, I think hierarchical clustering gives a more accurate grouping of the countries into the correct continent. Table 4 list the continents for the final clusters.

Table 4 Continents for final clusters

Cluster	Continent
0	Oceania (Australia and pacific) and Antarctica
1	Africa and Asia (Middle East)
2	Europe
3	South America
4	Asia (East Asia, South Asia and Southeast Asia)
5	North America
6	Oceania (remote Pacific islands)